Youngjoon Cheon

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Homepage ◇ **Google Scholar**

RESEARCH INTERESTS

- **Accelerator Memory Systems & Heterogeneous Architecture**: Architectures for expanding usable memory capacity and bandwidth in data-intensive workloads
- **Hardware–Software Co-Designed Memory Management**: Workload-aware mechanisms for data placement, movement, and address translation
- **Accelerator Execution & Coordination**: Exploring accelerator execution and coordination bottlenecks arising from interactions among workloads, hardware, and system software

EDUCATION

Korea University Sep. 2024 – Aug. 2026
M.S. in Electrical and Computer Engineering, GPA: 3.94/4.5 Seoul, South Korea
Thesis: *Segmentation-based Address Translation in GPUs for Accelerating LLM Inference*
Advisor: Prof. Jeongseob Ahn

Ajou University Mar. 2018 – Aug. 2024
B.S. in Software and Computer Engineering, GPA: 3.77/4.5 Suwon, South Korea
Minor in Mathematics

RESEARCH EXPERIENCE

Graduate Research Assistant Sep. 2024 – Aug. 2026
Computer Systems Laboratory, Korea University *Advisor: Prof. Jeongseob Ahn*

- **HW/SW Co-Design for LLM Address Translation [CAL'2026]** Jan. 2025 – May. 2026
  - Identified KV-cache-induced TLB contention that evicted reused model-weight translations during long-context LLM inference
  - Co-designed vLLM weight placement and a GPU segmentation table to bypass TLB translation for static, read-only weights
  - Reduced L2 TLB miss rate from 91.5% to 17.0% and achieved 2.51× IPC relative to paging in Accel-Sim
- **Intra-GPU Memory Tiering for KV-Cache Scaling** Apr. 2024 – Dec. 2024
  - Designed an HBM–LPDDR hierarchy with separate controllers, a unified address space, and layer-granular KV-cache migration
  - Integrated Ramulator with GPGPU-Sim to model distinct HBM/LPDDR timing behavior
  - Identified simulator-induced SM contention and built a DMA copy-engine model to decouple migration from computation

Undergraduate Research Intern Sep. 2022 – Aug. 2024
Computer Systems Laboratory, Ajou University *Advisor: Prof. Jeongseob Ahn*

- **Heterogeneous xPU–PIM Execution for LLM Inference** Jul. 2023 – Mar. 2024
  - Partitioned Hugging Face autoregressive decoding to offload memory-bound attention over the KV-cache to emulated PIM while retaining QKV projections and feed-forward layers on the xPU
  - Built a multi-GPU PIM emulation testbed and profiled inter-device overheads with NVIDIA Nsight Systems
  - Doubled memory capacity and batch size with 3.4% higher end-to-end latency than the single-GPU baseline

· **System Software & Parallel Programming Studies** Sep. 2022 – Jun. 2023

- Analyzed Data Access MONitor (DAMON) subsystem internals in the Linux kernel
- Investigated CUDA and parallel programming models for GPU accelerator research

PUBLICATIONS

Youngjoon Cheon, Yunho Oh, and Jeongseob Ahn, "Towards Segmentation-based Address Translation for LLM Inference," *IEEE Computer Architecture Letters* (**CAL**), 2026. [**Paper**]

AWARDS AND HONORS

Academic Excellence Scholarship (30-50% Tuition Waivers) Fall 2019, Fall 2022, Fall 2023
Ajou University

TEACHING EXPERIENCE

Graduate Teaching Assistant, Korea University

· **Digital Systems and Design Laboratory** Fall 2024, Fall 2025

- Conducted weekly lectures on digital logic and guided hands-on Verilog/FPGA synthesis labs

· **Computer Architecture** Spring 2025, Spring 2026

- Developed a RISC-V assembler programming assignment (**GitHub link**)

TECHNICAL SKILLS

Languages	Verilog, C/C++, CUDA, Python, Assembly, Shell
Frameworks	PyTorch, vLLM
Simulators & Tools	GPGPU-Sim, Accel-Sim, Ramulator, DRAMSim2, Nsight Systems/Compute, Vivado, Git

PROFESSIONAL SERVICE

Artifact Evaluation Committee Member

· ACM Symposium on Operating Systems Principles (SOSP 2026)
· IEEE International Symposium on High-Performance Computer Architecture (HPCA 2026)